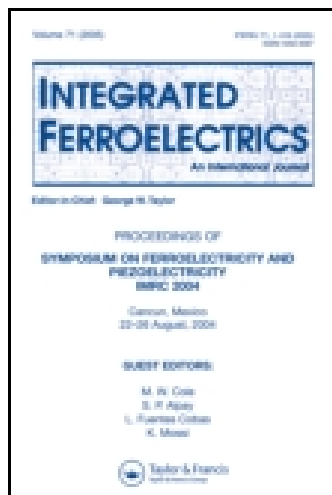




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ELECTRONIC STRUCTURE OF α AND β -PHASE OF POLY (VINYLIDENE FLUORIDE)

J. C. LI^a, C. L. WANG^a, K. YANG^a, X. Y. WANG^a,
M. L. ZHAO^a & J. L. ZHANG^a

^a School of Physics and Microelectronics, State Key Laboratory of Crystal Materials, Shandong University, Jinan, 250100, P.R. China
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Electronic Structure of α and β -phase of Poly (Vinylidene Fluoride)

J. C. Li,* C. L. Wang, K. Yang, X. Y. Wang, M. L. Zhao,
and J. L. Zhang

School of Physics and Microelectronics, State Key Laboratory of Crystal Materials,
Shandong University, Jinan 250100, P.R. China

ABSTRACT

The pseudopotential methods have been employed to investigate the electronic structures of α and β -phase poly (vinylidene fluoride). In the energy band structure of β -phase PVDF, a large dispersion of the valence band is found only along the chain direction, which agrees well with the angle-resolved photoemission experiments and implies the quasi-one-dimensional nature of β -phase PVDF. The large dispersion of the LUOM is caused by H *s*, C₂ *s p* state mainly. The energy band structure of β -phase exhibits anisotropy, while isotropy is found in the energy band of α -phase PVDF.

Keywords: PVDF; electronic structure; first principles calculation

In recent years, a great deal of attentions have been focused on typical ferroelectric polymers, poly (vinylidene fluoride), (PVDF) and its copolymer [1, 2]. Experimentally, It has been observed four forms: α , β , γ and δ phase in PVDF, of which only β phase is ferroelectrics [3, 4]. The cell of α -phase contains two molecular chains, which exhibit tans-gauche-tans-gauche' (TGTG') conformation [5]. The molecules of β -phase exhibit all-trans conformation. The spontaneous polarization of β -phase PVDF is along b axis of lattice cell, and about 0.065 C/m² [5]. Though many differences between α and β phases PVDF have been found in experiments and theoretical researches, there are still open problems, i.e. what is the difference of the electronic structure between the two phases. Solving the problem is helpful for deeply understanding the physical properties of the ferroelectric polymers. The purpose of this work is to investigate the electronic structures of α and β phase, and try to find the difference of the electronic structure between the two phases.

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*Corresponding author. E-mail: lijichao@sdu.edu.cn

In this work, all the calculations were carried out using DFT plane-wave pseudo-potential method implemented in the CASTEP code [6]. The exchange-correlation potential was treated with the generalized gradient approximation (GGA) in PW91 form. Vanderbilt type ultrasoft was used. A plane-wave energy cutoff of 370 eV has been adopted throughout, which was found to be sufficient large to yield convergent calculations. H(1s), C(2s, 2p) and F(2s, 2p) were treated as valence in our calculations, and the $(5 \times 3 \times 5)$ Monkhorst-Pack grid was used for the Brillouin zone (BZ) integration. The lattice parameters are taken from Ref. (7) for α -phase, $a = 4.96 \text{ \AA}$, $b = 9.46 \text{ \AA}$, $c(\text{fiber axis}) = 4.62 \text{ \AA}$, and $\beta = 90^\circ$, its space group is $P2_1/c$. The atomic fractional coordinates are H (0.366, 0.096, 0.760), F (0.218, 0.070, 0.276), C_1 (0.238, 0.183, 0.825) and C_2 (0.308, 0.186, 0.150). For β -phase, its space group is C_{m2m} . The lattice parameters are $a = 8.58 \text{ \AA}$, $b = 4.91 \text{ \AA}$ and $c(\text{fiber axis}) = 2.56 \text{ \AA}$. The atomic coordinate for the β -phase are H(0.105, -0.124 , 0.0), F(0.126, 0.355, 0.5), C_1 (0.0, 0.0, 0.0) and C_2 (0.0, 0.174, 0.5). For the two phases, the C_1 and C_2 denote the C atom bonding with H atom and C atom bonding with F atom, respectively.

The partial density of state (DOS) of α and β -phase are shown in Fig. 1. From the figure, it is noted that the DOS of the two phases is similar to each other, no obvious difference between them is found. In the range from 0 to -5 eV , the DOS of H s (shown in Fig. 1(a)) and $C_1 p$ (shown in Fig. 1(c)) have nearly the same character, this means that there is hybridization between H s and $C_1 p$ electrons. From the Fig. 1(d) and (e), it is easy to find that there are several peaks arising from $C_2 s P$ state in the range of $-5 \sim -10 \text{ eV}$. It is also found in Fig. 1(g) that there exist two peaks of the DOS of F p state from -5 eV to -10 eV . This indicates a hybridization exists between F p and $C_2 s, p$. Comparing Figs. 1(d), 1(e) and 1(f), it is found that F s state electrons take part in the hybridization with s and p electrons of C_2 atom in the range of $-20 \sim -25 \text{ eV}$. There are two high peaks in the range of $0 \sim -5 \text{ eV}$ in Fig. 1(g), this indicates that some p electrons of F atom do not take part in the hybridization between F and C_2 atoms. It is easy to understand this, because one F atom has seven valence electrons, and one C atom has four valence electrons, what's more, some of valence electrons of C_2 take part in the hybridization with valence electrons C_1 atom. From Figs. 1(b) and 1(d), it can be seen that, in the range from -10 eV to -15 eV , $C_1 s$ and $C_2 s$ have nearly the same valence density of state, this suggests that the covalent bond between C_1 and C_2 is mainly the result of hybridization between $C_1 s$ and $C_2 s$ electrons.

The band structure of β -phase obtained from our calculations is plotted in Fig. 2(a). The band structure is typical an insulating system with an energy gap of about 6.15 eV , which compares better with experimental value of 6.5 eV [8] than the previous calculation of 5.9 eV [9]. From the figure, it can be seen that the energy dispersion of the valence band is very small along a and b axes, which are perpendicular to the PVDF chain (denoted as $K_\perp$). Especially the highest occupied molecular orbital is almost dispersionless along the $K_\perp$

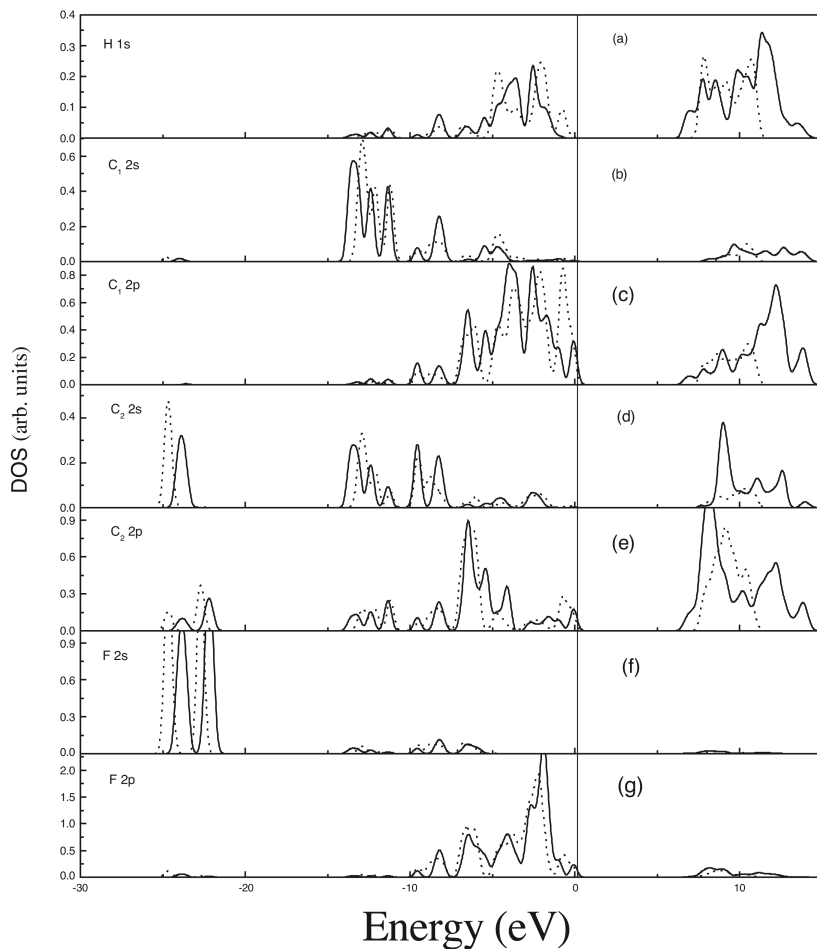


Figure 1. Partial DOS of α and β -phase. Solid line denotes DOS of β -phase, broken line denote DOS of α -phase. (a)–(g) show the partial DOS of the H s , C₁ s , C₁ p , C₂ s , C₂ p , F s and F p state, respectively.

direction. On the contrary, large dispersion can be seen along the c -axis, which is the PVDF chain direction (denoted as $K_{||}$). A similar situation is found for the lowest unoccupied band (LUMO), i.e., the band dispersions along $K_{||}$ are much larger than those along $K_{\perp}$. This has been found in the angle-resolved inverse photoemission experiment [10, 11]. Thus, both experiment and theory demonstrate that the interactions between different PVDF chains are rather weak and verify the low-dimensional electronic properties of PVDF. The theoretical amplitudes of the band dispersions along $K_{||}$ and $K_{\perp}$ are 1.89 and 0.3 eV, which agree very well with the experimental values 1.5 and 0.2 eV, respectively [10].

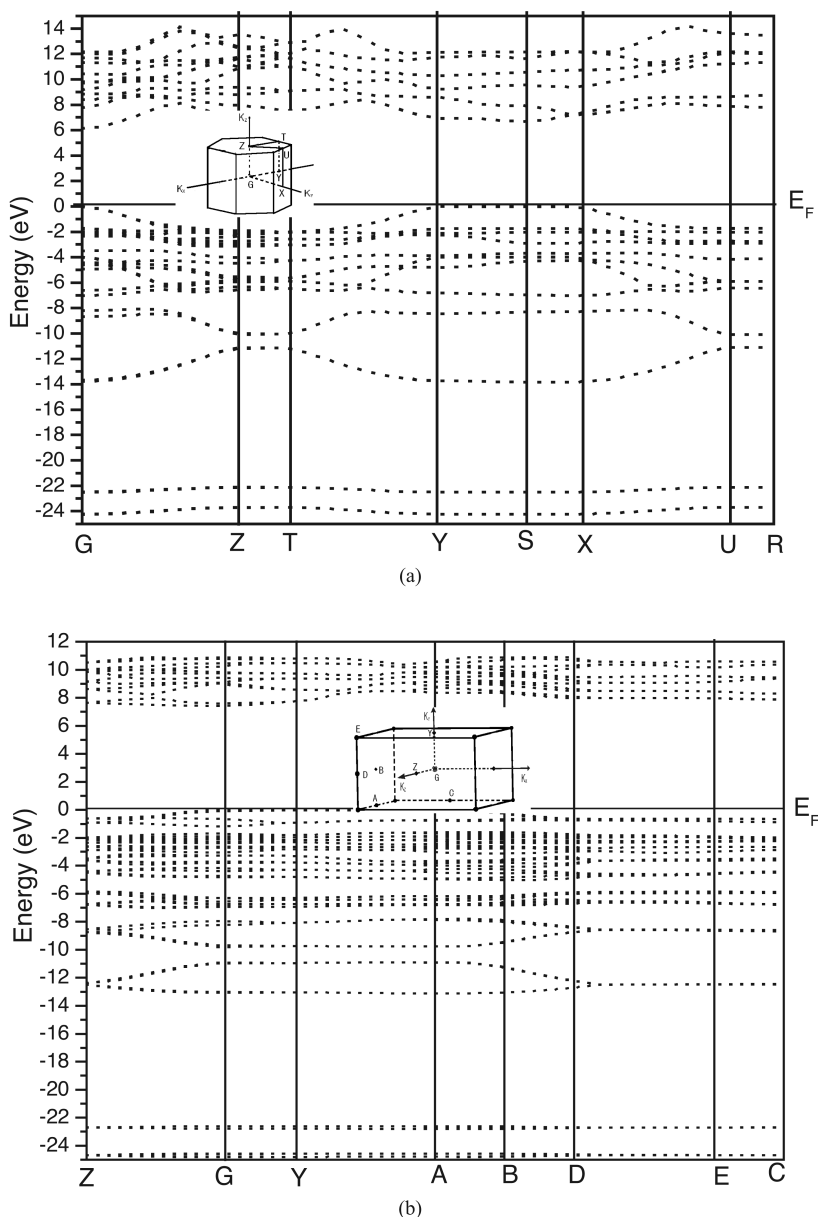


Figure 2. The energy band structure of β and α -phase PVDF. Fig. 2(a) and 2(b) correspond to β and α -phase, respectively.

Compared the Figs. 1 and 2, it can be seen that the large dispersion of the LUOM is caused by H s , C₂ $s p$ state mainly, and the F atom almost has no contribution to the large dispersion.

Comparing the Figs. 2(a) and 2(b), it is noted that there is dramatic difference between them. An obvious anisotropy was found in the energy band structure of β -phase, however, for the α -phase, it is obvious that all the energy bands are almost parallel to each other, this indicates that the band structure of α -phase exhibits isotropy. The electronic density in (001) surface of cell of the two phases is shown in Fig. 3. From the Fig. 3(a), it can be seen that

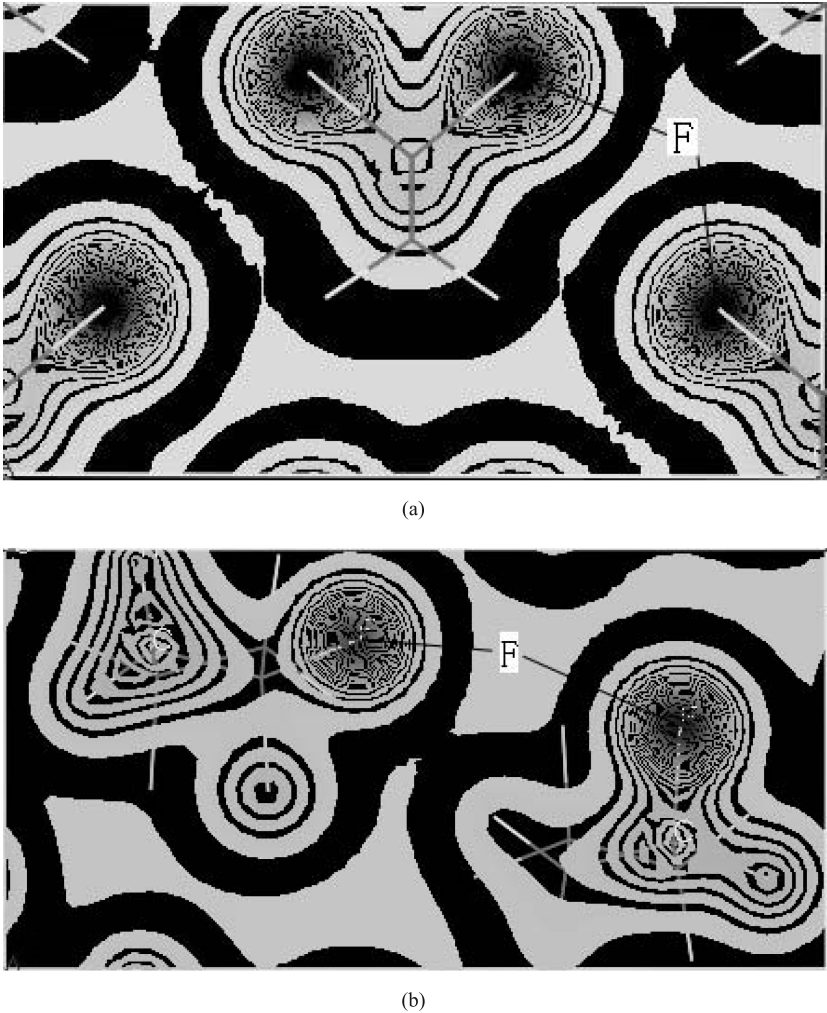


Figure 3. The electronic density in (001) surface of the cell of α and β -phase PVDF.

the electronic density of β -phase exhibits anisotropy, and the b axis is special. Like the energy band structure, the character of the electronic density of α -phase (shown in Fig. 3(b)) is isotropy. Since the bonding of atoms in the two phases is the same, the difference of the energy band structure and electronic density between the two phases should be caused by the difference of molecular conformation and space group. So the anisotropy of the energy band and electronic density may be important for the ferroelectric properties of β -phase.

In summary, the electronic structures of α and β -phase are investigated in detail. By analyzing the partial DOS, It is found that the electronic hybridization among the atoms is similar for the two forms. There is a hybridization between $H\ s$ and $C_1\ p$ state, the hybridization between $C_1\ s$ and $C_2\ s$ contributes to the bonding between C_1 and C_2 atoms, the hybridization exists between $C_2\ s, p$ and $F\ s, p$ electrons. For the β -phase, a large dispersion of the valence band is found only along the chain direction, which agrees well with the angle-resolved photoemission experiments, and implies the quasi-one-dimensional nature of β -phase PVDF. And the large dispersion of the LUOM is caused by $H\ s, C_2\ s\ p$ state mainly. The energy band and electronic density of β -phase exhibit anisotropy, however, the isotropy is the character of the energy band and electronic density of α -phase.

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